Rewarded for hostility? How voter engagement on social media shapes politicians' polarising communication in Sweden and Finland

A proposed PhD project within "A Divisive Spiral? Elite Communication, Voter Engagement, and Algorithmic Amplification of Affective Polarization" (PI Hanna Bäck) · Asad Ikram · PA2026/2879

1. Problem, aim and research questions

The project's premise is a spiral: politicians use polarising rhetoric, voters reward it with engagement, platforms amplify what is engaging, and politicians learn from the response. The voter side has experimental support. The elite side, whether engagement actually changes how representatives speak, has hardly been tested, and testing it requires measuring polarising communication in ways that can be compared across platforms, parliament, languages and modes. My dissertation would address the project's second work package, on how citizen engagement shapes representatives' communication strategies, and supply the measurement it shares with the first and third.

RQ1. Can out-party hostility, its target party and its discrete emotions (anger, fear, contempt) be measured equivalently in Swedish and Finnish, on social media and in parliamentary debate, and in text and speech? **RQ2.** Do representatives' hostile posts receive more engagement than their other posts, and towards which parties? **RQ3.** When hostility is rewarded, do representatives become more hostile, online and in parliament?

2. Previous research and gap

Affective polarisation, dislike and distrust of opposing partisans, has risen in many democracies (Iyengar et al. 2019). In multiparty systems it is directed at several parties to different degrees (Reiljan 2020; Wagner 2021). Elites are a plausible cause. In a Swedish survey experiment, factual tweets carrying an in- or out-party source cue raised affective polarisation, mainly among strong partisans (Bäck et al. 2023); threat raises affective polarisation through anger rather than fear (Renström, Bäck and Carroll 2023). Online, posts about the out-group attract more engagement (Rathje et al. 2021), users rewarded for outrage express more of it later (Brady et al. 2021), and ranking algorithms amplify some political content more than other content (Huszár et al. 2022). Among politicians, hostility on Twitter is reciprocated, more so among men (Wendsjö, Bäck and Kokkonen 2025).

Three gaps follow. First, measurement of elite hostility in the Nordic parliaments rests on dictionaries: new members are more negative towards other parties in the Storting (Røed, Bäck and Carroll 2025), and speech about the Sweden Democrats has become less negative (Calvo, Bäck and Carroll 2024). These studies capture valence rather than discrete emotions, identify the target party by name counts, and call for "alternative ways to validate measures of elite sentiment" (Røed et al. 2025); Calvo (2025) notes that language models validated by human coding would be the next step. Second, the reward side has been studied on Twitter alone, while engagement now happens on Facebook, Instagram and TikTok, which differ in format and ranking. Third, a comparison between parliament and social media is only meaningful if the measure works the same in both. Text is shaped by the medium (Sen et al. 2021), and in a methods paper I show that pooling machine-labelled sentiment across platforms without testing equivalence can reverse the sign of the aggregate on nearly a quarter of days (Ikram and Haig 2026).

3. Theory and hypotheses

I treat representatives as strategic communicators who learn from feedback. Engagement is an immediate, public and quantified signal of what activates supporters; following reinforcement-learning accounts of online expression (Brady et al. 2021), rewarded rhetoric should be repeated. Social identity theory explains why hostility is rewarded: attacks on an out-party affirm the in-group, and anger is the emotion that most strongly links threat to polarisation (Renström et al. 2023). Parliament differs: its norms of debate and its audience of colleagues should restrain the transfer.

H1. A representative's posts expressing out-party hostility receive more engagement than their other posts, and hostility expressed as anger is rewarded more than hostility expressed as fear. **H2.** The reward is larger for hostility towards ideologically distant and electorally competing parties. **H3.** Representatives whose hostile posts are rewarded become more hostile in later posts. **H4.** The effect carries over into parliamentary speech, but more weakly. **H0 (RQ1).** Without correction, labels of hostility are not equivalent across arenas, and uncorrected comparisons overstate the gap between social media and parliament.

4. Data and design

Parliament: plenary transcripts and recordings from the Riksdag and Eduskunta open data, covering at least the Swedish elections of 2018, 2022 and 2026 and the Finnish elections of 2019 and 2023, aligned with the corpora built in related projects. **Social media:** posts by members of both parliaments and the party accounts on Facebook, Instagram, X and TikTok, with engagement and, where available, reach, through official research access (the Meta Content Library, the TikTok research interface and data access for vetted researchers under Article 40 of the EU Digital Services Act). **Design:** RQ2 is estimated at the post level with representative fixed effects, so each person is compared with themselves, controlling for topic, format, timing and audience size. RQ3 uses a representative-by-week panel in which unusually high engagement on hostile posts predicts later hostility, online and in parliament, with campaign periods and leadership changes as comparison points. Because engagement is observational, the causal role of algorithms and of messages for voters is left to the project's experiments, to which this work contributes real, measured posts as valenced stimuli, as Bäck et al. (2023) recommend.

5. Measurement and validation

A codebook defines out-party hostility, its target party and its discrete emotions, with Swedish and Finnish examples. Two large language models label every unit in the original language, and speech is transcribed and checked on a sample. Native-speaker coders label a stratified gold sample for each combination of arena, language and mode. Following the protocol in Ikram and Haig (2026), I estimate how much score variance comes from arena, language and model, test measurement invariance and differential item functioning across them, and link scores where they differ. Dictionary scores from earlier studies serve as a benchmark, so the new measure can be compared with the old one.

6. Dissertation format

A compilation thesis of three articles and an introductory chapter: (1) a measurement article on equivalent multilingual measures of elite hostility across parliament and social media; (2) the engagement-reward article (RQ2); (3) the feedback article (RQ3), with a co-authored contribution to the project's experiments where it fits.

7. Timeline

Year 1: courses in theory, methods and research ethics; codebook; data access; Swedish courses. **Year 2:** gold coding and article 1. **Year 3:** article 2 and stimuli for the experiments. **Year 4:** article 3 and the introductory chapter.

8. Ethics and data management

Posts and speeches by elected representatives are public, but they reveal political opinions, a special category under GDPR; processing would rest on the public interest in research and on data made public by the data subjects, with data minimisation, pseudonymised storage of any private users' replies, and approval from the Swedish Ethical Review Authority where required. Platform terms and the conditions of research access will be followed, and code and codebooks shared.

9. Expected contribution

The project gains a validated measure of polarising elite communication in two languages and across arenas, and the first test of whether engagement rewards hostility and feeds back into it in multiparty systems. The measurement approach also applies to any comparison of political text across media.

References

Bäck, H., Carroll, R., Renström, E. and Ryan, A. (2023). Elite communication and affective polarization among voters. *Electoral Studies* 84: 102639.

Brady, W. J., McLoughlin, K., Doan, T. N. and Crockett, M. J. (2021). How social learning amplifies moral outrage expression in online social networks. *Science Advances* 7(33).

Calvo, E. (2025). *Old Foe, New Friend?* PhD thesis, Lund University.

Calvo, E., Bäck, H. and Carroll, R. (2024). Debating the populist pariah. *Political Research Quarterly* 77(3): 950–961.

Huszár, F. et al. (2022). Algorithmic amplification of politics on Twitter. *PNAS* 119(1).

Ikram, A. and Haig, E. (2026). Is the platform part of the measurement? A protocol and simulation study for cross-platform equivalence of machine-labelled policy sentiment. arXiv:2610.16757.

Iyengar, S., Lelkes, Y., Levendusky, M., Malhotra, N. and Westwood, S. J. (2019). The origins and consequences of affective polarization in the United States. *Annual Review of Political Science* 22: 129–146.

Rathje, S., Van Bavel, J. J. and van der Linden, S. (2021). Out-group animosity drives engagement on social media. *PNAS* 118(26).

Reiljan, A. (2020). 'Fear and loathing across party lines' (also) in Europe. *European Journal of Political Research* 59(2): 376–396.

Renström, E. A., Bäck, H. and Carroll, R. (2023). Threats, emotions, and affective polarization. *Political Psychology* 44(6).

Røed, M., Bäck, H. and Carroll, R. (2025). Who polarizes parliament? *Party Politics* 31(1): 137–147.

Sen, I., Flöck, F., Weller, K., Weiß, B. and Wagner, C. (2021). A total error framework for digital traces of human behavior on the web. *Public Opinion Quarterly* 85(S1): 399–422.

Wagner, M. (2021). Affective polarization in multiparty systems. *Electoral Studies* 69: 102199.

Wendsjö, A., Bäck, H. and Kokkonen, A. (2025). A male hostility spiral? Polarized communication among political elites on social media. *Journal of Politics*.

Word count, including references: 1,500.